MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8660

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal

See figure 2

Mounting position: Any

Weight: 27 ounces (765.5 grams) nominal

ABSOLUTE RATINGS: F = 110 MHz

Parameter: Unit:	Ef Vac	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ib mAdc	Pg1 W	Pg2 W	Pp W	Anode core and seal T °C	tk sec (min)	Cooling --- (Note 1)
C teleg:	6.0 ±5%	3,000	-250	350	750	1.0	12	1,500	250	180	---
Class B or AB:	6.0 ±5%	3,000	---	400	900	1.0	12	1,500	250	180	---
TEST CONDITIONS:	6.0	2,000	Adj	300	250	---	---	---	---	300	Note 2

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	9.0	11.0	Aac
(B)1261	Electrode voltage (grid)	-		0.65	II	Ec1	-48.0	-62.0	Vdc
1256	Electrode current (screen)	-		0.65	II	Ic2	0	-10.0	mAdc
1266	Primary grid emission (control)	9	Ec1/Pg1 = 1.0 W; p = g2 = k = 0 Vdc; Eavg (reverse) = 25 V; t = 60	0.65	II	Isg1	---	-10	µAdc
1266	Primary grid emission (screen)	9	Ec2/Pg2 = 12 W; p = g1 = k = 0 Vdc; Eavg (reverse) = 50 V; t = 60	0.65	II	Isg 2	---	-25	µAdc
---	Ion current	3	Eb = -67.5 Vdc; Ec2 = 300 Vdc; Ec1/Ic2 = 20 mAdc	0.65	II	Iz	---	10	µAdc
(B)2204	Linear amplifier power output and distortion	7	Class AB amplifier; Eb = 2,900 Vdc; Ec2 = 225 Vdc; Ec1/Ibo = 300 mAdc; Eg1/Ib = 700 mAdc; RL = 2,200 ±100 ohms; Rg = 1,000 ohms (max); anode loaded tank O = 10 to 15	---	---	Po 3rd IM 5th IM	1,000 -35 -40	--- --- ---	W dB dB
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance (grounded cathode connection)	-		---	---	Cgp Cin Cout	--- 75.0 10.8	0.03 88.0 12.8	pF pF pF
)1372	Current division	-	Eb = 500 Vdc; Ec2 = 325 Vdc; Ec1 = -200 Vdc; egk/ib = 3.0 a	---	---	egk ic1 ic2	--- --- ---	20 50 300	v ma ma
	Quality conformance inspection, part 3								
---	Life test	8	Group C; linear amplifier power output and distortion, except Eg1/Po = 1,100 W (useful); t = 500 hours	---	---	---	---	---	---
---	Life test end point:								
3)2204	Linear amplifier power output and distortion	-		---	---	Po 3rd IM 5th IM	1,000 33 38	--- --- ---	W (useful) dB dB
---	Vibration (noise)	4, 5	Ebb = 2,500 Vdc; Ec2 = 400 Vdc; Ec1/Ib = 250 mAdc; Ef = 6.0 Vac or Vdc; Rp = 1,000 ohms; Accel = 10 G peak (min); F = 28 to 500 Hz; ascending sweep only	---	---	Ep	---	30	Vac

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Vibration (noise) end point:								
Ⓑ 1261	Electrode voltage (grid)	-		---	---	Ec1	-48.0	-62.0	Vdc
Ⓑ ---	Shock, specified pulse (long duration)	4, 6	Eb = 2,000 Vdc; Ec2 = 400 Vdc; Ec1 = -200 Vdc; Shock = 11 ±2 ms; Accel = 50 G peak (min); total impacts = 9	---	---	---	---	---	---
Ⓑ ---	Shock, specified pulse (long duration) end point:								
1261	Electrode voltage (grid)	-		---	---	Ec1	-48.0	-62.0	Vdc
---	Stability	4, 7	Eb = 2,900 Vdc; Ec2 = 225 Vdc; Ec1/Ibo = 300 mAdc; t = 90	---	---	ΔIb	---	30	mAdc

NOTES:

1. In all cases of operation, forced-air cooling is required and the maximum temperature ratings should not be exceeded. An air-system socket, such as the EIMAC SK-800 series, and air director (chimney) SK-806, or equivalents, shall be used. The cooling table is based on operation of the tube in an SK-800B socket with SK-806 chimney, with air flowing in a base-to-anode direction, with cooling air at 50°C maximum, and for operation at frequencies below 30 MHz. Additional air cooling may be required at higher frequencies. Airflow must be applied before or simultaneously with electrode voltages (including the heater), and may be removed simultaneously with them. When long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

Anode dissipation	Sea level		10,000 feet	
	Airflow (cfm)	Approx press drop ($In.H_2O$)	Airflow (cfm)	Approx press drop ($In.H_2O$)
500 W	7.5	0.04	11	0.06
1,000	17.5	0.23	24	0.21
1,500	34	0.60	45	0.80

2. In all electrical tests involving application of heater voltage, the use of an air-system socket and chimney is allowable, and forced-air cooling is permitted in order to maintain temperatures within the allowable maximum.

3. This test shall be the first performed after the holding period. The tube shall be connected as an ion gage, with the specified potentials applied, and the highest initial reading shall be taken as the test value. Ion current is read with a microammeter in series with the negative anode lead. Except for Ef, all voltages shall be applied simultaneously, with an automatic current regulator controlling Ic2 to the value specified.

Ⓑ 4. This test shall be performed yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

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'OTES: -Continued

5. Each tube under test shall be subjected to one sweep cycle in each of the three axes, X, Y, and Z. One sweep (28 to 500 Hz, ascending only) shall be covered in 6 to 15 minutes. The specified voltages shall be applied during the test, using the basic circuit shown on figure 3. Tubes found to electrically oscillate for causes other than vibration shall not be tested nor rejected on this test. Each tube shall be vibrated for 60 seconds at the frequency which gives the maximum vibration (noise) output voltage in each of the three axes. If at the end of the 60 seconds the vibration output voltage is increasing, the vibration shall be continued until there is no further increase. The tubes shall not show noise voltage output in excess of the maximum limit specified, except one intermittent short shall be allowable during this test. In addition to reading noise voltage on the specified VTVM, or equivalent, a permanent recording shall be made using a good quality recorder to produce a plot of noise voltage versus frequency. Noise voltage amplifiers used with the recorder shall have a ±1 dB frequency response over the range to be measured and the overall recording equipment shall be capable of fast response in order to show sharp noise voltage spikes resulting from internal tube resonances or other phenomenon. Prominent noise peaks indicated on the recording shall be individually investigaged by fixed-frequency operation, and the 60-second operation shall be made at the frequency of highest noise as so selected.

6. Each tube under test shall be subjected to the specified acceleration in the X, Y, and Z axes, with three shocks in each axis. Tubes showing any permanent shorts, or more than one temporary short during the test, shall be rejected. The applied shock shall be an approximate half-sine wave motion with the duration measured at the zero-axis level.

7. When the stability test is performed, the same sample of tubes shall be used for the stability test and the linear amplifier power output and IM distortion test, with the stability test performed first. The same test circuit shall be used for both tests. The absolute value of the 3rd IM shall never be greater than the absolute value of the 5th IM.

8. The life test shall be considered as destructive. All voltage and current meters shall be calibrated for ±2 percent accuracy or better as installed in the equipment. The circuit used need not be identical to that used for the linear amplifier power output and IM distortion test. Prior to the performance of the life test, the tubes used shall have met the requirements of the linear amplifier power output and IM distortion test.

9. The test circuit of figure 1 shall be used. The input voltage shall be varied to obtain the average inverse voltage and the resistor R shall be varied to obtain and maintain the forward power for the specified time. All unused tube elements shall be grounded. The wattmeter shall be capable of reading over the range from dc to at least 600 Hz. Loading of the transformer is permissible.

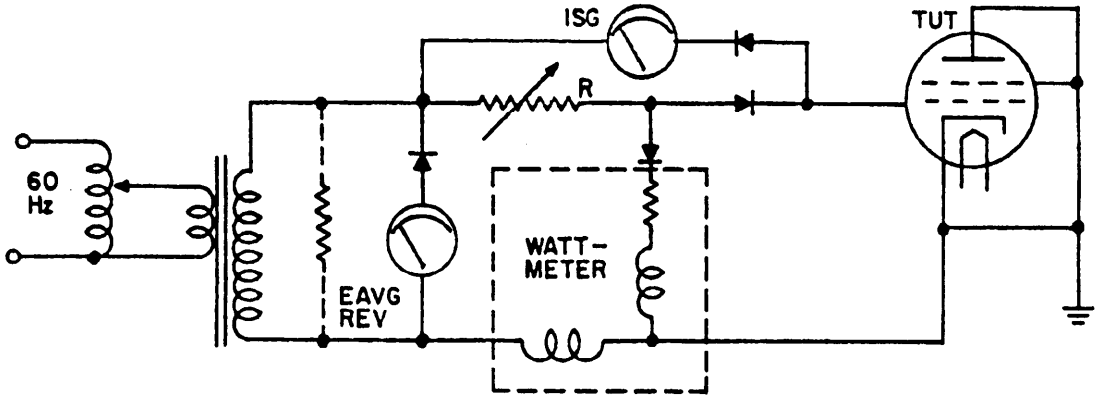


FIGURE 1. Test circuit.

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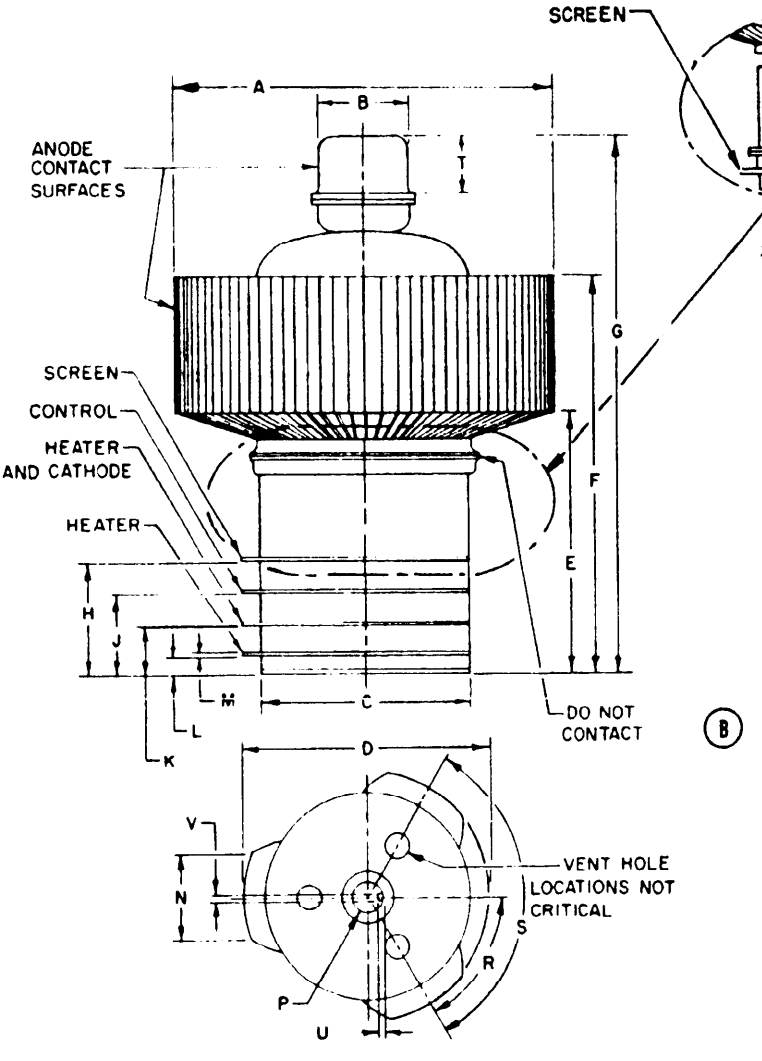
User activities:
Navy - AS, OS, MC, CG

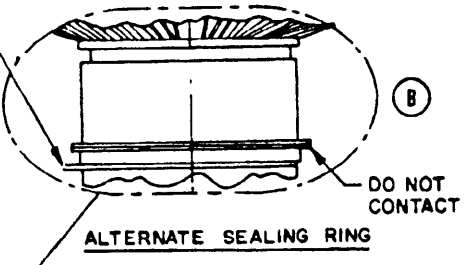
Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
C	1.870 (47.50)	1.900 (48.26)
G	4.600(116.84)	4.800(121.92)
H	.965 (24.51)	.988 (25.10)
J	.690 (17.53)	.710 (18.03)
K	.415 (10.54)	.435 (11.05)
L	.140 (3.56)	.165 (4.19)
M	.020 (.51)	.030 (.76)
P	.314 (7.98)	.326 (8.28)
U	.025 (.64)	.048 (1.22)
V	.045 (1.14)	.070 (1.78)
Quality conformance inspection, part 3 (see note b)		
A	3.335 (84.71)	3.365 (85.47)
B	.807 (20.50)	.817 (20.75)
D	2.250 (57.15)	2.300 (58.42)
E	2.195 (55.75)	2.380 (60.45)
F	3.410 (86.61)	3.550 (90.17)
N	.700 (17.78)	.800 (20.32)
R	55°	65°
S	115°	125°
T	.470 (11.94)	.530 (13.46)

NOTES:
a. Dimensions H, J, K, L, and M shall be maintained within .100 inch (2.54 mm) from the outer peripheral edge of the contact tabs.
b. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 2. Outline drawing of electron tube type 8660.

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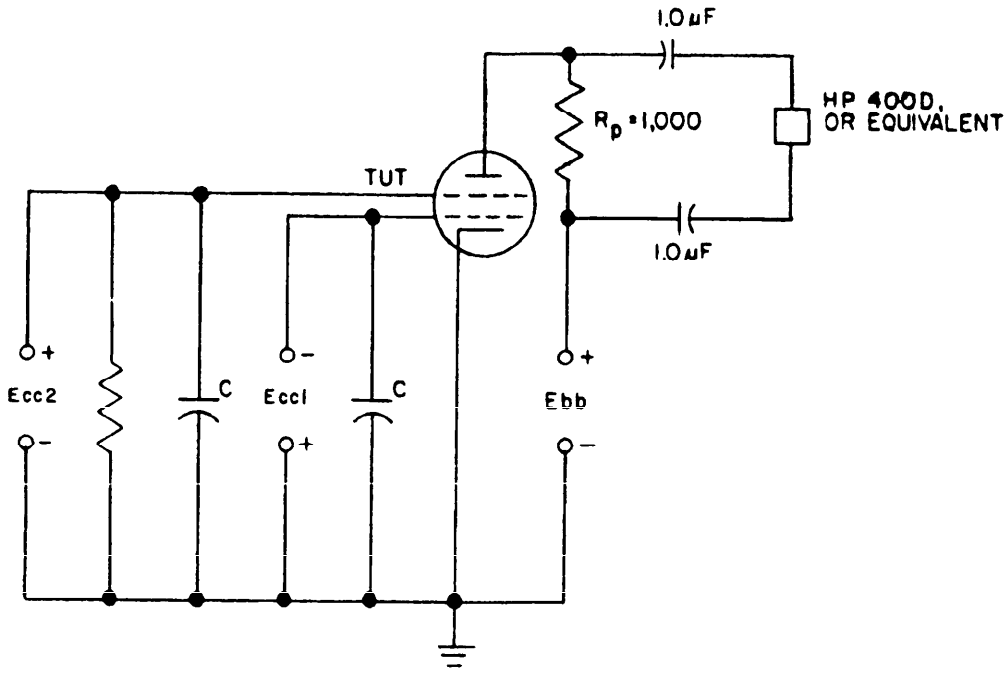


FIGURE 3. Basic circuit for use in vibration (noise) test.

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